

Draft - expert review pending

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Draft translation - not yet human-reviewed / مسوده ترجمه

Algebraic Reasoning

Algebra is not a new subject; it is the number work of Unit 2 made reusable. This unit walks the classic road into it: patterns first (because a pattern's rule is an expression waiting to be written), then equations and inequalities (because a rule plus a known total is a solvable claim), and finally the translation loop that turns fares, budgets and rate cards into algebra and back into decisions.

Unit learning outcomes

By the end of this unit you should be able to:

- Find the growth step of a pattern and write its rule as an expression with a named variable.
- Solve two-step linear equations with a substitution check, and read the equals sign as a balance.
- Solve linear inequalities, including the sign flip on negative multiplication, and round whole-number answers in the direction the situation demands.
- Translate a fare, budget or pricing situation into an expression or equation, solve it, and interpret the result back into the situation.

Prerequisite knowledge

Unit 2 (especially operations, decimals and the operation-choosing flowchart) and Unit 1's four-step problem-solving strategy.

In this unit

1. [Topic 3.1 - From patterns to algebra: variables and expressions](#) - the road from matchstick patterns to rules written with variables.
2. [Topic 3.2 - Linear equations and inequalities](#) - the balance method, the check, and the range answers.
3. [Topic 3.3 - Using algebra in problem solving](#) - the translate-solve- interpret-decide loop on real rate cards.

How to use this unit

Work the topics in order. Keep a pencil moving: every worked example hides its conclusion one line below where you should stop and try. The practicum task in each topic collects real rate cards and patterns; ERQ-5 will ask for yours, so keep the notes dated. Glossary terms are underlined; hover or tap them for the Urdu definition.

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